Pythagorean Theorem and Coordinate Distance

Grade 8 / Pre-Algebra

Hypotenuse from legs, missing legs, coordinate distance, and multistep composites. The first two problems are the warm-up; after that the four methods are mixed, so decide for yourself which one a problem calls for. For every problem: sketch the right triangle, label what you know, and write $a^2 + b^2 = c^2$ with c on the *longest* side before you compute. Round only at the very end, and only when the problem says to.

1. A loading ramp rises 5 ft straight up over a horizontal run of 12 ft. How long is the ramp surface itself?

Answer: ________________ ft

2. A 17 ft ladder leans against a vertical wall with its base 8 ft out from the wall. How high up the wall does the top of the ladder reach?

Answer: ________________ ft

3. On a city map drawn on a coordinate grid, the library sits at $(2, 3)$ and the pool at $(7, 15)$. Each grid unit is one block. How far apart are they in a straight line, measured in grid units?

Answer: ________________ units

4. A rectangular garden measures 24 m long by 10 m wide. Rosa walks from one corner to the opposite corner two ways: along two sides of the garden, or straight across on the diagonal path. How many meters shorter is the diagonal walk?

Answer: ________________ m

5. Two boats leave the same dock at the same time. One sails 6 km due north; the other sails 9 km due east. How far apart are the boats when they stop? Round to the nearest tenth of a kilometre.

Answer: ________________ km

6. A drone flies in a straight line from the point $(-4, -2)$ to the point $(5, 10)$ on a coordinate grid. How long is its flight path in grid units?

Answer: ________________ units

7. A ranger station stands at $(6, 8)$. A straight trail runs along the x-axis, so every point on the trail has the form $(x, 0)$. Find *both* points on the trail that are exactly 10 units from the station.

Answer: ________________

8. A kite is flying at the end of 65 ft of taut, straight string anchored to the ground. The kite is 60 ft above the ground. How far is the anchor from the spot on the ground directly beneath the kite?

Answer: ________________ ft

9. A 13 ft ladder reaches 12 ft up a wall. Someone bumps it and the base slides out until the top only reaches 5 ft up the wall. How much farther from the wall is the base of the ladder now?

Answer: ________________ ft

10. A delivery van drives a closed route on a coordinate grid: warehouse $(1, 2)$ to stop A $(7, 10)$, then straight down to stop B $(7, 2)$, then straight back to the warehouse. How long is the whole route in grid units?

Answer: ________________ units

11. A cell tower stands at $(3, -2)$. A straight power line runs along the vertical line $x = 8$, so every point on it has the form $(8, y)$. Find *both* points on the power line that are exactly 13 units from the tower.

Answer: ________________

12. **Synthesis challenge.** A storage crate is a rectangular box measuring 120 cm long, 90 cm wide, and 80 cm tall. A straight pole is to be laid inside it, running from one bottom corner to the opposite top corner. What is the greatest length a pole can have and still fit? (Use the theorem twice: first across the bottom of the crate, then up to the far top corner.)

Answer: ________________ cm